

ECE 1508: Applied Deep Learning

Course Logistics

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Welcome to ECE 1508!

Happy to see you in ECE 1508

Applied Deep Learning

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Where and When?

- *Tuesdays* at **1:00 PM till 3:00 PM** at **MC-252**
- *Thursdays* at **1:00 PM till 3:00 PM** at **BL-205**

Tutorials: There are tutorial sessions

- TBA

Teaching Team

- Faeze Moradi – *Project Supervision*
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Quercus and Piazza

We got a *Quercus* page

- You have been automatically enrolled
- Also you got registered at the *Piazza* page
 - ↳ You can login through the *Quercus* page
- We though use the *Course Page* to share the course materials

Please! Feel free to ask questions on Piazza!

What Do We Learn?

Simple: we are going to learn **Deep Learning!**

You may wonder how do we learn it? Well! in 3 steps

- **Step 1: Preliminaries of Machine Learning**
 - We try to get what is the problem in Machine Learning
 - We understand what Deep Learning is
 - We get to know Neural Networks and their Deep version
 - We understand why really Deep Neural Networks work

By the end of this step, we know in theory

- How Deep Learning works
- What **we need to learn** if we want to build a Deep Learning model

This motivates us to get to the next step!

What Do We Learn?

*Simple: we are going to learn **Deep Learning**!*

You may wonder how do we learn it? Well! in 3 steps

- **Step 2: Neural Networks: MLPs, CNNs and Sequence Models**
 - We now get to know the details of each architecture
 - How we can implement them
 - What kind of challenges we deal with when we implement them
 - What are the standard techniques to overcome these challenges

This is the major part of the course. As we get over this part

- You can consider yourself a **mid-level expert** in **Deep Learning**
- You are able to build a suitable **Deep Learning model** for a **given problem**
- You are able to **implement** what you need **from scratch**

No worries! We all get there for sure! 😊

What Do We Learn?

Simple: we are going to learn *Deep Learning!*

You may wonder how do we learn it? Well! in 3 steps

- **Step 3: Advances in Deep Learning**
 - We study few selected advance topics in Deep Learning
 - We see how easily we follow the topics using what we learned in *Step 2*

By the end of this part

- You believe that you're mid-level expert in Deep Learning
- You get to know how to train yourself for more advanced projects

Checkout the course [syllabus](#)

How Do We Get Trained?

There are three learning components in the course

- *Assignments*
 - ↳ We solve **four sets** of assignments
 - ↳ **No need to say** that they are the **best thing** to understand the course!
 - ↳ And, of course we do **lots of programming** in there!
 - ↳ Each assignment will be solved in Tutorial **after the deadline**
 - ↳ You have **two weeks** time for each assignment
 - ↳ Submission by **deadline at 11:59 PM**
 - ↳ **No late** submission is accepted!

Attention!

Assignments make almost **half** of the **course mark**; so, they need effort! They are **a part** of the **learning process**, **not supplementary**!

- ↳ You must use AI **responsibly**!
- Midterm Exam
- Final Project

How Do We Get Trained?

There are three learning components in the course

- Assignments
- *Midterm Exam*
 - ↳ We will have **one written exam** in the **middle of semester**
 - ↳ Questions that **can be solved by hand**, so **no programming** in the exam
 - ↳ We just evaluate our **understanding** of **fundamental concepts**
 - ↳ Exam is on **February 26, 2026 at 1:00 PM**
- Final Project

How Do We Get Trained?

There are three learning components in the course

- Assignments
- Midterm Exam
- *Final Project*
 - ↳ The **most interesting** part of the course
 - ↳ We build **groups** of size **three**
 - ↳ **Each group** chooses a topic:
*some **design problem** OR a **research-oriented** topic*
 - ↳ **Each group** submits a proposal by **Week 5**
 - ↳ **Each group** briefs us about the progress by **Week 10**
 - ↳ **Each group** accomplishes their project
 - ↳ We meet all in a **seminar** where the groups present their projects

*Read **the course page** for details*

Course Calendar

Week #	Date	Notes	Posted	Deadline
1	Jan 05 - Jan 09			
2	Jan 12 - Jan 16		Assignment 1: Basics	
3	Jan 19 - Jan 23			
4	Jan 26 - Jan 30		Assignment 2: FNNs	Assignment 1: Basics
5	Feb 02 - Feb 06			Project: Flyer and Proposal
6	Feb 09 - Feb 13			Assignment 2: FNNs
7	Feb 16 - Feb 20	Reading Week-- No Lectures		
8	Feb 23 - Feb 27	Midterm Exam on Feb 26	Assignment 3: CNNs	
9	Mar 02 - Mar 06			Assignment 3: CNNs
10	Mar 09 - Mar 13		Assignment 4: Sequence Models	Project: Progress Briefing
11	Mar 16 - Mar 20			
12	Mar 23 - Mar 27			Assignment 4: Sequence Models
13	Mar 30 - Apr 03			
14	Apr 06 - Apr 10	Examination Time -- No Lectures	Project: Presentation	Project: Final Report and Source Codes

- You could check it out [here](#)

In Person Lectures!

The course is **in-person** and . . .

. . . *by Regulations the attendance is mandatory in in-person courses!*

attendance
MATTERS

Recording is uploaded after each lecture

- They are mainly for **recap** and/or **following some missed lectures**
- I **strongly** advise to **attend** the lectures
 - You don't want to miss my **performance** 😊
 - **Board and I** are **not** captured in videos, it's **only** the **slides and my voice**

Lots of Programming in Python

We are going to do lots of programming in Python

- ↳ Some basic knowledge in Python is needed
 - ↳ **Don't run away** if you **haven't**, you only need a bit of *more efforts*
- ↳ We just need **basics**, we will learn all packages that we need
 - ↳ *Matplotlib, NumPy, Scikit-Learn, Pandas*
 - ↳ But our main toolkit in Python will be **PyTorch** that we learn in detail
- ↳ **Don't mistake!** We do **not** learn *based on PyTorch!*
 - ↳ We learn to implement pretty much everything *from scratch*
 - ↳ But, we need to know **PyTorch** as well, since it is the *professional toolkit*

No Major Prerequisites

The course is *self-containing* meaning that

you will learn all background you need!

We assume that we all have some *basic math* in mind

↳ linear algebra, calculus, and probability theory

But we review *whatever we need* from these topics *whenever needed!*

There are some other *related courses*, the most important one is

- ECE 1513: Introduction to Machine Learning

- ↳ Totally fine (actually *suggested*) if you *had it before or have it right now*

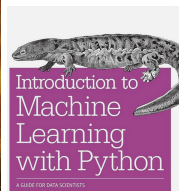
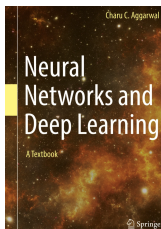
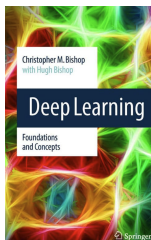
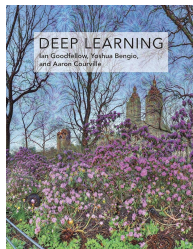
- ↳ It's focusing on the *general Machine Learning* not *Deep Learning*

- ↳ We are going to *study Deep Learning and Neural Networks*

- ↳ So, the two courses can be seen as *complements*

- ↳ You *could* also have *only one of them*

Textbooks



All materials *are provided in the course*. It's however *good to know* some texts!

- Goodfellow et al., can be accessed online [at this link](#)
- New textbook by Bishops is available online [here](#)
- Also, you may take a look at *NN and Deep Learning* [here](#)
- For *PyTorch* the best resource is its own [tutorials](#)

Terms and Conditions!



The instructor keeps the right reserved for himself to modify the slides

- *last minute before the lecture 😊*
- *after the lecture has been given*
 - *Typically happens due to typos*

The instructor keeps the right reserved for himself to deliver the lecture-notes

- *in form of mini-batches 😊*

Date and Signature

Introducing Glum

Glum does not buy my words! e.g.,

- + *Well! We know Python, you think you can teach us Deep Learning!*
- *Sure! Let's try!*

So, please excuse me if I explain things sometimes in too much detail! I need to convince Glum!



No such thing as a stupid question!

Did you know that we got [a Wikipedia page](#) on this?

- ↳ Trust me! Your question will **never** sound stupid!
- ↳ **If you don't ask**; then, **I need to ask**!
 - ↳ **Interaction** is the best tool to avoid getting bored!

Any Questions? 😊